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Antecedents and innovation outcomes of not-invented-here syndrome – Examination at individual, managerial, and organizational levels

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Abstract

A negatively shaped attitude-based bias towards external knowledge—the not-invented-here (NIH) syndrome—may constitute a significant obstacle for effective adoption of external knowledge and impair organizational innovativeness. The goal of this study is to shed light to the question how organizational and managerial features drive individual NIH, and how NIH connects further to individual innovativeness. Analysis of survey data collected from 93 employees reveals a joint effect of person-organization fit and leader-member exchange on NIH, and a negative association between NIH and innovative work behavior. Contrary to expectations, person-organization fit and perceived innovativeness of organizational culture do not have joint effects on NIH. The findings contribute to the development of a theoretical model of the interplay between individual-, managerial- and organizational-level antecedents and consequences of the NIH syndrome.

1 | INTRODUCTION

Organizations aiming for innovations and growth often open their innovation processes for external knowledge. The goal is to develop the organization's own knowledge base while sharing the related resources and risks with others (e.g., Chesbrough, 2006; Huizingh, 2011; Husted & Michailova, 2002). From employees, the practice of external knowledge sharing often necessitates changes in the existing ways of working. Therefore, it sometimes is met with resistance. Previous studies have documented that a negatively shaped attitude-based bias towards externally generated knowledge, labeled as not-invented-here (NIH) syndrome, may constitute a significant obstacle for effective external knowledge sharing at the individual level (e.g., Antons & Piller, 2015). Relatedly, previous research has linked NIH syndrome to project failures and a decline in firm performance (e.g., Herzog & Leker, 2010).

While the research interest towards the NIH phenomenon has been abundant since the seminal work of Katz and Allen (1982), a recent review finds the extant understanding of the antecedents and underlying mechanisms of the NIH construct still rather limited (Antons & Piller, 2015). The limited understanding of factors potentially influencing employees' NIH attitudes is a notable shortcoming because individual employees play a salient role in succeeding to meet an organization's innovation goals (Afsar et al., 2015). Aiming to narrow this gap, this study joins the stream of research aiming to better understand the NIH syndrome at the level of individual employees (e.g., Antons et al., 2017; Burcharth & Fosfuri, 2015; Herzog & Leker, 2010). The research question guiding this study is, *how organizational and managerial features drive individual NIH syndrome, and how NIH syndrome connects further to individual innovativeness*.

Firstly, this study examines the assumption of a direct negative association between an employee's NIH attitudes and individual-level innovative work behavior (IWB) and, thereby, seeks to shed additional light to possible attitude-behavior linkages in employee innovativeness. While the conceptual development of the NIH construct has been grounded on the premise of a higher level of NIH attitudes predicting a lower level of employee innovativeness (e.g., Antons & Piller, 2015), thus far, the empirical evidence of the associations between NIH syndrome and various behavioral outcomes at the level of individual employees is inadequate.

Second, the goal of this study is, first, to contribute to the development of a more nuanced view of the organizational conditions that have the potential to shape NIH syndrome. Contextual qualities have been found to influence individuals' attitudinal outcomes at the workplace (e.g., Dewettinck & van Ameijde, 2011; Edgar & Geare, 2005). Among the many variables addressed in the research into organizational behavior, person-organization fit (P-O fit) has been identified as a significant predictor of various employee attitudes (Cable & Judge, 1996; Leung et al., 2011). P-O fit represents the compatibility between an individual and the organization and, in most studies, has been conceptualized as the congruence between an individual's personal values and the values represented in the organizational culture (Kristof-Brown et al., 2005). P-O fit has been linked to employees' innovation-related outcomes such as innovative work behavior (e.g., Afsar, 2015; Saether, 2019) and creativity (e.g., Saraç et al., 2014; Seong & Choi, 2019). However, less is known about the possible predictive role of P-O fit for employees' innovation-related attitudes such as NIH syndrome. This study maintains that a better understanding of the possible *interplay between P-O fit and NIH syndrome* has the potential to add detail to the conceptual development of the NIH construct. However, the relationship between P-O fit and NIH syndrome is likely to be contingent to social and interpersonal contexts (see Seong & Choi, 2019). Firstly, the type of the organizational culture is likely to play a significant role in P-O fit – NIH syndrome relationship. It is intuitively and theoretically plausible that if an employee's values are aligned with the organizational values, the congruence is likely to predict a lower level of NIH attitudes, specifically, *when the employee perceives the organizational culture as innovative*. Second, not only the broader organizational environment such as organizational culture, but also the more immediate workplace context reflecting the managerial features has the power to influence employees' attitudes (see Erdogan et al., 2002). One particularly salient factor is the relationship between an employee and his/her immediate supervisor, conceptualized as leader-member exchange (Dienesch & Liden, 1986; Graen & Uhl-Bien, 1995). Previous studies have associated leader-member exchange (LMX) with employees' innovative outcomes such as innovative behavior (Basu & Green, 1997; Scott & Bruce, 1994) and creativity (e.g., Pan et al., 2012; Tierney, 2015). Furthermore, LMX is shown to influence employee creativity in tandem with P-O fit (Seong & Choi, 2019). While the theoretical underpinnings and extant findings, thus, imply that *the interplay between P-O fit and LMX quality may be relevant* for *innovation-related attitudes such as NIH syndrome*, the question still warrants further scrutiny.

The aims of the study are met through theoretical and empirical examination of the possibly compensating or reinforcing roles of organizational-level and managerial-level features in influencing employees' NIH attitudes. Specifically, this study addresses relevant research investigating the roles of P-O fit, organizational culture, and LMX as possible antecedents of employees' workplace attitudes and behaviors in general, and individual innovativeness in particular. Data collected from 93 employees in an innovating organization provides empirical insight into the relevant relationships.

This study has the potential to contribute to the literature on NIH syndrome in several ways. By unfolding the possible joint effects of the organizational and managerial features on NIH syndrome, this paper adds detail to the theoretical development of the antecedents of the NIH syndrome and draws attention to the NIH construct specifically at the individual level. For the wider innovation management literature, the contribution of this study lays in revealing the mechanisms of how innovation-related attitudes occur, and how can they be supported at managerial and organizational levels to promote subsequent innovativeness. Thereby, this study answers the calls of Anderson et al. (2014) and Hammond et al. (2011) to explore possible mechanisms through which employee innovation can be driven.

The remainder of this paper is structured as follows. First, the association between NIH syndrome and IWB is addressed. This discussion is followed by consideration of the possible roles of P-O fit, the perceived innovativeness of organizational culture, and LMX in shaping employees' NIH attitudes. Drawing from existing theorizing, the overall research framework and the hypotheses are developed. Next, the data collection and methods are discussed, and the empirical results are presented. Finally, the findings of the empirical study are discussed, and theoretical and practical implications are outlined. The paper concludes with a discussion of the study's limitations and suggestions for future research directions.

2 | THEORETICAL FRAMEWORK AND HYPOTHESES

2.1 | The level of innovative work behavior as a direct consequence of NIH attitudes

Innovation-generating companies often choose to adopt external ideas and information to complement internal resources and leverage opportunities outside the organizational boundaries (Chesbrough, 2006; Huizingh, 2011). By engaging in external knowledge sharing, the companies have the potential to advance organizational learning and improve their strategic network position. However, knowledge-sharing beyond the traditional organizational boundaries poses challenges that range from the outflow of business-critical information to missed inflow of value-adding external knowledge caused by biased acceptance within the organization (Burcharth & Fosfuri, 2015; Husted & Michailova, 2002; Ritala et al., 2015). At the individual-level, an attitude-induced decision-making bias towards externally generated knowledge is identified as a significant obstacle for effective external knowledge sharing (Antons et al., 2017; Gesing et al., 2015; Greco et al., 2019; Katz & Allen, 1982). The prejudice labeled as not-invented-here (NIH) syndrome refers to the presence of a profound bias against ideas and knowledge that are generated outside the focal organization (Gesing et al., 2015; Grosse et al., 2012). The subjective perception of externality in an employee's mind is of relevance; externality does not involve external parties only but may refer to another business unit or area of expertise within the own organization (Antons & Piller, 2015). Employees' NIH attitudes may lead to siloed approaches where the organization lacks awareness of the lessons available from others' perspectives. The consequences can be project failures (Herzog & Leker, 2010) and a decline in company performance (Katz & Allen, 1982).

At individual level, the theorizing around the NIH construct has evolved from the idea that employees' NIH attitudes are negatively associated with their innovative actions (e.g., Antons et al., 2017; Greco et al., 2019). However, the literature on the interplay between an individual's NIH attitudes and his/her actual behavior is mainly anecdotal (see Antons et al., 2017). We provide empirical evidence of the intuitively and theoretically plausible negative association between an employee's NIH attitudes and his/her innovative behavioral outcomes by focusing especially on the relationship between NIH and innovative work behavior (IWB). IWB was chosen as the outcome variable as it has emerged as a particularly important concept in the research into employee innovativeness (De Jong & Den Hartog, 2010; Janssen, 2001). It is characterized as comprising three forms of behavior which represent different stages of the innovation process—idea generation, promotion, and implementation—and is seen to constitute behaviors that are specially targeted to benefit the organization (Janssen, 2001; Scott & Bruce, 1994). The present study suggests that it is plausible that an employee, who seeks and adopts external knowledge with an open-minded attitude is more likely to actively generate, promote and implement novel ideas with an aim to develop the existing ways of working than an employee with a negatively biased attitude towards external ideas and knowledge. Drawing from the theorizing on the NIH construct and on IWB, this study, thus, maintains that an employee with a high level of NIH attitudes is less likely to pursue IWB than an employee with a low-level NIH.

Hypothesis 1. *Not-Invented-Here syndrome is negatively associated with innovative work behavior.*

2.2 | The contingent effects of person-organization fit on NIH syndrome

Individuals have been found to differ in their NIH attitudes (e.g., Burcharth & Fosfuri, 2015; Herzog & Leker, 2010; Mehrwald, 1999), and the suggested causes that would explain the preference for one's own knowledge instead of adopting others' ideas include a lack of trust towards the quality of the external knowledge, professional pride referring to the thought that it is more prestigious to create new knowledge instead of reusing existing ideas, and a strong group affiliation, resulting in a 'them versus us' differentiation (Husted & Michailova, 2002; Katz & Allen, 1982; Mehrwald, 1999). Furthermore, prior studies have documented factors such as organizations' institutionalized socialization practices (Burcharth & Fosfuri, 2015), collaboration governance mode (Gesing et al., 2015), the number of projects one is involved in (Grosse et al., 2012), company size (Agrawal et al., 2010) and the standing or status of the external party (Reitzig & Sorenson, 2013) as antecedents of the NIH syndrome. Still, the *antecedents and underlying mechanisms of the NIH at individual level* are not yet fully understood (Antons & Piller, 2015).

Person-organization fit (P-O fit) has often been addressed in the management and organizational behavior literature as a salient predictor of employees' attitudinal outcomes (Cable & Judge, 1996; Kristof-Brown et al., 2005). P-O fit is defined as congruency between an employee's values and the cultural value system of an organization (for a meta-analysis of P-O research, see Kristof-Brown et al., 2005). The theoretical premise for expecting P-O fit to be conducive to employees' desired workplace attitudes and behaviors has been grounded in the notion that when employees hold values that match the values of the organization, the employees identify better with the organization and are more motivated to maintain the employment relationship and engage in extra-role behaviors (Kristof-Brown et al., 2005; O'Reilly et al., 1991). In line with the theorizing, previous research has demonstrated significant positive association between P-O fit and, as an example, job satisfaction and organizational commitment (Chatman, 1991; Cooper-Thomas et al., 2004; O'Reilly et al., 1991).

In the research on NIH syndrome, the possible role of P-O fit in predicting employees NIH attitudes has, however, largely been neglected. This is a shortcoming, firstly, as previous research suggests that employee-organization value congruence has the potential to shape employee innovativeness. Prior studies have associated P-O fit with innovative employee outcomes such as innovative work behavior (e.g., Afsar, 2015; Huang et al., 2019; Saether, 2019) and creativity (e.g., Saraç et al., 2014; Seong & Choi, 2019). Klein and Sorra (1996), for example, suggested that employees are more willing to accept and implement innovations when the innovations are consistent both with employees' individual values and organizational values. Second, in a conceptual study, Antons and Piller (2015) suggested that NIH attitudes may serve, among others, a *value-expressing function* in individuals' lives by offering a means for expressing, activating, and affirming one's central values (Antons & Piller, 2015).

Drawing from the previous findings and theorizing, the present study maintains that person-organization value congruence has the potential to affect employees' NIH attitudes. We further propose, however, that the association between P-O fit and NIH syndrome is assumably not straightforward but contingent to contextual characteristics. This proposition is grounded on the notion that, whereas previous research has found strong positive associations between P-O fit and many desired employee outcomes such as organizational commitment and extra-role behavior (Kristof-Brown et al., 2005), P-O fit may not always be conducive for change-oriented outcomes (Kristof, 1996). Keeping in mind that NIH syndrome is suggested to derive from an individual's need for stability and security (Antons & Piller, 2015) as adopting external knowledge, typically, involves disruption in the status quo and increases the level of uncertainty (Grosse et al., 2012) it is of relevance to consider *whether specific contextual characteristics may have the potential to turn the effect of P-O fit in favor of more change-oriented attitudes such as lower levels of the NIH syndrome.*

2.2.1 | The role of the organizational culture in P-O fit-NIH syndrome association

Generally, the type of organizational culture is found to have important implications for employee attitudes and behaviors (O'Reilly, 1989). Organizational culture reflects the values of the organization, is communicated through norms, artifacts, and is observed in behavioral patterns (Schein, 1992). One commonly accepted conceptualization assesses organizational culture along three dimensions—innovative, bureaucratic, and supportive—each characterized with distinguishing organizational values (Wallach, 1983). *An innovative organizational culture* is described as dynamic, result-oriented, entrepreneurial, and creative (Wallach, 1983). Not surprisingly, previous studies have generally established a positive relationship between the innovativeness of the organizational culture and employee innovative outcomes (for a meta-analysis, see Büschgens et al., 2013), thereby connecting organizational features and individual attitudes and behavior. The findings of Herzog and Leker (2010) suggest that organizational culture plays a role in employees' NIH attitudes also. They found that levels of NIH attitudes were lower among employees working in business units with open innovation culture than among their peers working in units characterized by closed innovation culture (Herzog & Leker, 2010).

Employees with a better P-O fit engage more in discretionary behaviors (Kristof-Brown et al., 2005). When looking at the nature of the discretionary behaviors, it feels intuitively probable that when employees perceive that the organizational culture supports, encourages, and even expects the employees to search and try out external knowledge and ideas to develop organizational innovativeness, the employees with high P-O fit would engage more in extra-role behaviors especially in this direction. In other words, *P-O fit is likely to predict a lower level of NIH attitudes, specifically when the employee perceives the organizational culture as innovative.* Thus, drawing from the previous findings and extending prior knowledge, we argue that employees' perceptions of the innovativeness of organizational culture (henceforth organizational innovativeness) is an important component in the association between P-O fit and NIH syndrome. It is plausible that P-O fit weakens the NIH attitudes particularly when the employees perceive the organizational culture, with which their personal values are aligned, as innovative. Accordingly, the following hypothesis is formulated:

> **Hypothesis 2.** *The joint effect of person-organization fit and organizational innovativeness on Not-Invented-here attitudes is negative, specifically, when organizational innovativeness is high.*

2.2.2 | LMX as a potential moderator in P-O fit – NIH syndrome association

In addition to organizational features, also managerial features may emerge as relevant for the P-O fit–NIH syndrome connection. Previous studies have argued that important work relationships such as leader-member exchange may change the effects of P-O fit on employee attitudes and behavior (e.g., Boon & Biron, 2016; Erdogan et al., 2002; Seong & Choi, 2019). *Leader-member exchange (LMX)* refers to the dyadic relationship between a supervisor and each of his/her subordinates (Dienesch & Liden, 1986). According to LMX theory, a leader forms high-quality, in-group relationships with a small number of subordinates, and low-quality, out-group relationships with the rest (Graen & Uhl-Bien, 1995). High-quality LMX is characterized by loyalty, commitment, support, and trust between the dyad members, and employees in high-quality relationships are given greater resources, decision latitude, and freedom. In contrast, a low-quality relationship is based merely on the formal job requirements and employment contract (Uhl-Bien & Maslyn, 2003).

LMX quality plays an important role in shaping many employee workplace outcomes and studies have associated high-quality LMX with higher job satisfaction (e.g., Rockstuhl et al., 2012), better work performance (e.g., Buch et al., 2016), and more citizenship behaviors, among other outcomes (e.g., Ilies et al., 2007). In line with the main body of LMX research, innovation management scholars have generally documented a positive association between LMX and employees' innovative outcomes such as IWB (Agarwal et al., 2012; Basu & Green, 1997; Scott & Bruce, 1994). Also, employee creativity has been positively linked to LMX (e.g., Pan et al., 2012; Tierney, 2015).

LMX quality is contingent to the mutual expectations and fulfillment of the exchanges between a leader and a follower (e.g., Bauer & Erdogan, 2015; Dienesch & Liden, 1986). When looking at the exchanges at different organizational levels, prior studies have suggested that LMX as a direct organizational exchange relationship would have the potential to facilitate the more distant organizational exchange relationship of P-O fit (Erdogan et al., 2002). Most previous studies have found an enhancing effect of LMX in associations between P-O fit and employee behavioral and attitudinal outcomes. Boon and Biron (2016) documented that P-O fit was positively associated with person-job fit when LMX was high but not when LMX was low. The results of Seong and Choi (2019) showed that LMX activated the significance of P-O fit towards creativity, and the results of Kim et al. (2013) revealed that high-quality LMX enhances the positive effects of P-O fit on perceived social exchange with organization. In turn, Erdogan et al. (2002) reported P-O fit to increase job and career satisfaction for employees specifically in low LMX relationships. The results of Jung and Takeuchi (2014) offer one explanation to the partially mixed findings by showing that national cultural features may explain part of the variation. Specifically, their study showed that that LMX moderated the positive relationships between P-O fit and job satisfaction, and organizational commitment, respectively, for Japanese employees but not for Korean employees (Jung & Takeuchi, 2014).

Adopting external knowledge and ideas into one's work is, for many employees, beyond the explicitly mentioned job obligations. Thus, the support, resources, and negotiation latitude that supervisors provide to employees in high-quality LMX relationships may encourage these employees to seek and try externally generated ideas and knowledge and introduce them to other organizational members (see Greco et al., 2019). Therefore, the positive effects of P-O fit in weakening NIH attitudes may be activated, specifically, when employees experience high quality LMX relationships. Consequently, we argue that high-quality LMX activates the negative effects of P-O fit on employee NIH attitudes. This leads to the third hypothesis of the study:

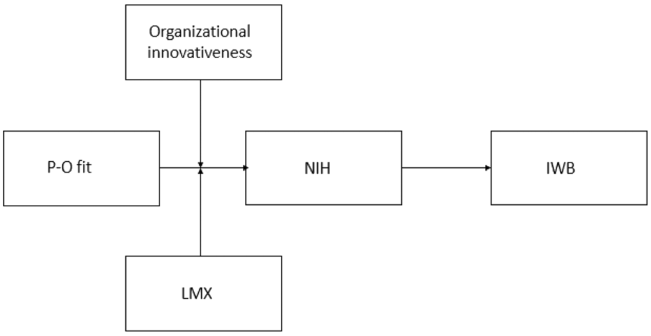

FIGURE 1 Hypothesized relationships between not-invented-here attitudes, innovative work behaviors, person-organization fit, the innovativeness of organizational culture, and leader-member exchange quality

> **Hypothesis 3.** *The joint effect of person-organization fit and leader-member exchange quality on Not-Invented-Here attitudes is negative, specifically, when leader-member exchange quality is high.*

Figure 1 summarizes the proposed framework of the associations between NIH syndrome, P-O fit, organizational innovativeness, LMX, and IWB.

3 | METHOD

3.1 | Sample and data collection

The hypotheses were tested with a sample of employees in product development jobs from a company in Finland. The company chosen for this study serves B2B customers in a knowledge intensive business field and operates both internationally and locally. Employee innovativeness is appreciated in the company in general, and in the studied division in particular. The data was collected in 2020 as part of a survey of innovation-related individual and organizational characteristics. The organization's HR department sent a link to a quantitative online survey to 131 employees in a specific division. Total of 93 employees responded the survey questions (response rate being 71%). The respondents were 50% female and 50% male. Of the respondents, 2% were under 25 years, 20% were 25–35 years, 20% were 35–44 years, 29% were 45–54 years, and 29% were 55–64 years old. The organizational tenure of the participants ranged from less than 1 year to 36 years (M = 12, SD = 10).

3.2 | Measures

A back-translation procedure was used to ensure coherence when translating the original (English) version of the measurement instruments into Finnish.

Not-Invented-Here (NIH) attitudes were captured by adopting a 9-item NIH syndrome scale developed and validated by Antons et al. (2017). Adopting a novel NIH syndrome measure allows us to contribute to the extant research on NIH syndrome not only theoretically but also methodologically. The respondents were asked to self-evaluate their attitudes towards external knowledge sharing by answering, on 5-point Likert scale, questions such as 'I network across different knowledge domains.' and 'Collaborating with other knowledge domains generates more overhead than benefit.' The Cronbach's alpha value of the construct measure was 0.76.

Employees' innovative work behavior (IWB) was measured using a 9-item scale developed and validated by Janssen (2001). The respondents were asked, using a 7-point Likert scale, to respond to questions such as 'How often do you pay attention to issues that are not part of your daily work?'; 'How often do you enlist support for innovative ideas?' and 'How often do you transform innovative ideas into useful applications?' The Cronbach's alpha was 0.94.

Person-organization (P-O) fit was assessed with a single-item measure where the respondents were asked, using a 5-point Likert scale, to evaluate the extent to which they perceive the values represented in the organizational culture are congruent with their own, individual values. While a single-item measure is not without its limitations, this measure was adopted as previous research indicates that employees' own estimations of their P-O fit serve as better predictors of their attitudes and behaviors than so called 'actual fit' measures that are calculated by matching organizational characteristics and personal values (Kristof-Brown et al., 2005).

To measure *organizational innovativeness*, the Organizational Culture Index (Wallach, 1983) was adopted. The Organizational Culture Index (OCI) is a measure of three dimensions of organizational value culture—bureaucratic, innovative, and supportive. It uses eight adjective descriptors to assess each of the three cultural facets, and the eight items measuring the innovativeness of the organizational culture were adopted for the analyses. Using a 4-point scale Likert scale, respondents were asked to assess how well the adjectives describe their organizations. Sample adjectives include 'Risk-taking', "Stimulating', and 'Creative.' The Cronbach's alpha was 0.65.

Leader-member exchange was assessed with the Finnish version (see Kauppila, 2016) of LMX-7 scale (Graen & Uhl-Bien, 1995) that is one the most frequently used and most recommended versions of LMX scales (Bauer & Erdogan, 2015). The respondents were asked to evaluate, on 5-point Likert-scale, statements such as: 'I know where I stand with my supervisor'; 'I usually know how satisfied my supervisor is with what I do', and 'I would characterize my working relationship with this my supervisor as highly effective'. The Cronbach's alpha = 0.89.

3.3 | Data analysis and results

SPSS Version 27 was used to analyze the data. Table 1 presents means, standard deviations, and correlations for the variables investigated in this study. From the correlations we see that NIH syndrome

TABLE 1 Descriptive statistics and correlation matrix

Variable	Mean (S.D.)	1	2	3	4
1. IWB	3.84 (1.00)				
2. P-O Fit	3.25 (0.65)	0.090			
3. Organizational m innovativeness	2.84 (.37)	−0.203	0.265*		
4. LMX	3.71 (.74)	−0.227*	0.148	0.311**	
5. NIH	2.33 (.48)	−0.318**	−0.119	0.015	−0.035

Abbreviations: Innovative, the innovativeness of organizational culture; IWB, innovative work behavior; LMX, leader-member exchange; NIH, not-invented-here; P-O Fit, person-organization fit.
*$p < 0.05$: **$p < 0.01$.

TABLE 2 Results of regression analysis

Variables	Dependent variable: NIH
1. P-O Fit	−0.106 (.080)
2. Organizational innovativeness	0.050 (.018)
3. P-O Fit * Organizational innovativeness	−0.177* (0.031)
F	1.463
R2	0.047
R2 Adj.	0.015

Notes: *$p < 0.10$: **$p < 0.05$, ***$p < 0.01$: Standard errors in parentheses, Abbreviations: NIH, not-invented-here; P-O Fit, person-organization fit.

TABLE 3 Results of regression analysis

Variables	Dependent variable: NIH
1. P-O Fit	−0.131 (.077)
2. LMX	−0.043 (.068)
3. P-O Fit * LMX	−0.195* (.097)
F	1.617
R2	.052
R2 Adj.	.020

Notes: *$p < 0.10$: **$p < 0.05$, ***$p < 0.01$: Standard errors in parentheses. Abbreviations: LMX, leader-member exchange, NIH, not-invented-here, P-O Fit, person-organization fit.

showed the expected negative correlation with IWB. Innovativeness of organizational culture showed a positive correlation with P-O fit and LMX, respectively. Analysis of the data showed no signs of multicollinearity.

Before proceeding to test the hypotheses, the data was checked for normality, and the independent variable (P-O fit) and the moderating variables (innovativeness of organizational culture and LMX) were centred. The data was nested in six teams, with the respondents being supervised by six different leaders. The preliminary analysis of the data showed no variance in NIH syndrome residing between the teams, suggesting that regression analysis methodology is suitable for analyzing the hypotheses. Nevertheless, after carrying out the regression analyses, HLM method (SPSS Mixed Models Linear) was adopted to re-test the hypothesized relationships. HLM results confirmed the findings of the regression analyses and, for parsimony, only these are reported.

Firstly, H1 proposed a negative linear relationship between employees' NIH attitudes and their IWBs. In line with the expectation, the regression analysis yielded a significant negative effect of NIH attitudes on IWB (−0.318, $p < 0.002$, F = 10.265, $R2 = 0.101$, Adjusted $R2 = 0.091$) which brought support for H1.

H2 suggested that the organizational innovativeness moderates the relationship between P-O fit and NIH syndrome. Table 2 presents the results of the regression analysis. From the results we can see that the interaction between P-O fit and innovativeness on NIH syndrome was significant just under $p < 0.10$ level ($p = 0.095$). A closer look at the associations at low (1 standard deviation below the mean) and at high levels (1 standard deviation above the mean), respectively, of

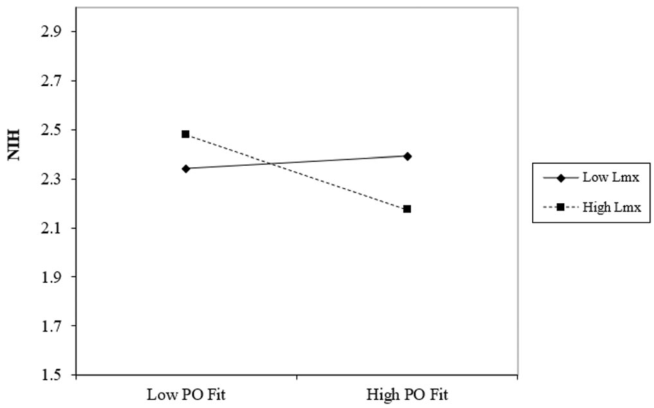

FIGURE 2 Interaction effect of leader-member exchange quality on the association between person-organization fit and not-invented-here attitudes

P-O fit and innovativeness was taken. A plot of the depicted interaction effect suggested that the association between P-O fit and NIH syndrome was negative both at a low and a high level of innovativeness, yet the interaction effect was not statistically significant. Consequently, H2 was not supported.

H3 proposed that LMX moderates the relationship between P-O fit and NIH. Table 3 presents the results of the regression analysis. Again, the interaction between P-O fit and LMX was approaching significance ($p = 0.065$) so the interaction effect was depicted in a similar manner as described above. Figure 2 presents the association between P-O fit and NIH syndrome at a low and at a high level of LMX (set at 1 standard deviation below and above the mean, respectively). Testing the simple slopes of the two lines indicated that when LMX was high, P-O fit had a significant negative association with NIH syndrome (simple slope $= -0.233$, $p = 0.04$) and when LMX was low, the relationship was not significant (simple slope $= 0.039$, $p = 0.70$). Thus, H3 was supported, bringing evidence for the moderating role of high-quality LMX in the P-O fit – NIH syndrome association.

4 | DISCUSSION

The open innovation and knowledge management literature is still in need for a more detailed understanding of mechanisms behind NIH syndrome and their management (Antons & Piller, 2015; Greco et al., 2019; Temel & Vanhaverbeke, 2020). This study demonstrates a direct negative relationship between NIH syndrome and IWB, thereby providing evidence of a direct, innovation-related attitudes-behavior linkage at the individual level. The findings highlight the salient role that the individual employees may play for the success of organizations' innovation plans and, furthermore, connect this study to prior research that has reported relative results at the organizational level. For example, Burcharth et al. (2014) documented a negative relationship between the average level of employees' NIH attitudes in the organization and the organization's open innovation practises.

Our results further show that LMX quality moderates P-O fit – NIH syndrome association in such a way that the relationship was negative only when LMX quality was high. The findings suggest that the employee-organization congruence of values, needs, and beliefs together with support and trust from the immediate supervisor can lead to less prejudiced employee attitudes towards external knowledge (Antons & Piller, 2015; Dienesch & Liden, 1986; Klein & Sorra, 1996). Seeking and trying out knowledge and ideas from outside can be time-consuming, arduous, and cost-intensive. While P-O fit ties employees to the organization and motivates for extra role behaviors (Kristof-Brown et al., 2005) it is usually the supervisor who can provide employees with avenues for searching for and experimenting with external knowledge by providing them with the needed resources and showing trust and support (c.f. Vila-Vázquez et al., 2020). The findings are in line with the study of Husted et al. (2012) suggesting that commitment-based mechanisms diminish employees' knowledge-sharing hostility. Likewise, our findings add to the research of Afsar et al. (2015), highlighting the important effects of the interplay between P-O fit and trust on employees' innovative outcomes.

Contrary to our initial expectations, moderation effect of innovative organizational culture could not be established between the relationship of P-O fit and NIH syndrome. However, this finding may be at least partially explained by the organizational setting of this study; the empirical examination was conducted in a single-organization context—and one that can be best described as innovation-driving. The participants of the study evaluated the perceived innovativeness of the organizational culture generally as relatively high (the mean was 2.84 in the scale from 1 to 4). However, given the fact that the expected interaction effect between P-O fit and perceived innovativeness of the organizational culture on NIH syndrome was approaching significance and was in the expected direction (negative), the results imply that the hypothesized interplay between P-O fit and organizational culture is possible. The research question warrants further research.

Taken together, our findings contribute to a more nuanced perspective of the process of external knowledge evaluation and adoption among individual employees. The findings of this study add detail both to the antecedents and consequences of the NIH syndrome and to the multilevel dynamics underlying NIH syndrome-IWB associations, thereby joining the research bringing together both the human and the organizational side of innovation (see Elmquist et al., 2009). For the wider innovation management literature, the contribution of this study lays in unfolding mechanisms of how individual-level innovation-related attitudes and behaviors occur and interrelate, and how can they be supported on managerial and organizational levels (see Greco et al., 2019).

5 | CONCLUSIONS

This study adds to the existing knowledge on the relationships between employees' innovation related attitudes and behaviors, specifically NIH syndrome and IWB. In particular, it contributes to earlier research by providing evidence on the interaction of P-O fit and LMX quality in relation to employees' NIH attitudes, thereby providing a more nuanced view on the individual and joint roles of factors residing at individual, managerial, and organizational levels.

For practitioners, this paper provides relevant insight needed to manage employee innovativeness in the workplace. Acknowledging NIH syndrome as an attitude, and IWB as a behavior, supports utilizing the managerial and organizational features to influence them. The findings remind practitioners of the importance of maintaining high-quality subordinate-supervisor relationships as a relevant means to stimulate employees' attitudes towards opportunities presented outside the organization. It advocates managerial attention from top-managers and encourages them to regularly evaluate the supervisor-subordinate relationships at different organizational levels and building channels of communication across the ladders of supervision in case challenges emerge in those relationships. Our study also sends an important message to recruiters to consider the individual value profiles of potential employees, and specifically the evaluate these against the organizational values.

The limitations of this study should be taken into consideration when assessing the findings. Firstly, the study is cross-sectional and limited in terms of organizational and national cultural settings. The sample size (while representative of the context) is small, which emphasizes the exploratory nature and limits generalizability of the findings. Future research should therefore employ longitudinal designs and larger quantitative data sets to better understand the dependencies between person-organization fit, innovative organizational culture, leader-member exchange quality, Not-Invented-Here attitudes, and innovative work behavior. Nevertheless, by drawing attention to these aspects and by demonstrating and explicating the relationships of these factors, this study provides steppingstones for such further research.

DATA AVAILABILITY STATEMENT

Research data are not shared.

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